Diffusion: The net movement of molecules from a region of their higher concentration to a region of their lower concentration along a concentration gradient until an equilibrium is attained.

Characteristics of diffusion

i. It is a passive process, hence no energy expenditure involved.
ii. It is independent of the living system.
iii. Diffusion is obvious in gases and liquids.
iv. Diffusion is rapid over a shorter distance but extremely slow over a longer distance.
v. The rate of diffusion is determined by temperature, concentration gradient and relative density.

Significance of diffusion in Plants

i. Gaseous exchange of O_2 and CO_2 between the atmosphere and stomata of leaves takes place by the process of diffusion. O_2 is absorbed during respiration and CO_2 is absorbed during photosynthesis.
ii. In transpiration, water vapour from intercellular spaces diffuses into atmosphere through stomata by the process of diffusion.
iii. The transport of ions in mineral salts during passive absorption also takes place by this process.



Diffusion for sterilization in surgical theatres

Surgical theatres must be free from germs to prevent infection during surgeries. A mixture of Formalin and Potassium permanganate produces enormous fumes which will kill all pathogens in an enclosed area. This method is known as **fumigation** and operates by diffusion.

2. Facilitated Diffusion

Cell membranes allow water and nonpolar molecules to permeate by simple diffusion. For transporting polar molecules such as ions, sugars, amino acids, nucleotides and many cell metabolites is not merely based on concentration gradient. It depends on,

i. ***Size of molecule***: Smaller molecules diffuse faster.
ii. ***Solubility of the molecule***: Lipid soluble substances easily and rapidly pass through the membrane. But water soluble substances are difficult to pass through the membrane. They must be facilitated to pass the membrane.

Types of Membrane Permeability

A solution is made up of solute particles dissolved in a solvent and the permeability of the above components depends on the nature of cell membranes, which is given below:

Impermeable: Inhibit the movement of both solvent and solute molecules. Example: Suberised, cutinised or liginifid cell walls.

Permeable: They allow diffusion of both solvent and solute molecules through them. Example: Cellulosic cell wall.

Semi permeable: Semi permeable allow diffusion of solvent molecules but do not allow the passage of solute molecule. Example: Parchment paper.

Selectively permeable: All bio membranes allow some solutes to pass in addition to the solvent molecules. Example: Plasmalemma, tonoplast, and membranes of cell organelles.

In facilitated diffusion, molecules cross the cell membrane with the help of special membrane proteins called transport proteins, without the expenditure of ATP.

There are two types of transport proteins present in the cell membrane. They are channel protein and a carrier protein.

I. Channel Protein

Channel protein forms a channel or tunnel in the cell membrane for the easy passage of molecules to enter the cell. The channels are either open or remain closed. They may open up for specific molecules. Some channel proteins create larger pores in the outer membrane. Examples: Porin and Aquaporin.

XIth_Botany_Chapter_11_EM.indd 235 05-01-2023 18:29:01